

```
In [11]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Observation:

We are importing three essential Python libraries for EDA:

- pandas: for data handling
- matplotlib.pyplot and seaborn: for creating visualizations

```
In [10]: df = pd.read_csv("train.csv")
```

Observation:

The Titanic dataset has been successfully loaded from the file "train.csv" into a pandas DataFrame called `df`.

```
In [3]: print("Shape of dataset:", df.shape)
print("\nData types and non-null counts:")
print(df.info())
print("\nSummary statistics:")
print(df.describe(include='all'))
```

Shape of dataset: (891, 12)

Data types and non-null counts:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

Summary statistics:

	PassengerId	Survived	Pclass	Name	Sex
count	891.000000	891.000000	891.000000	891	891
unique	NaN	NaN	NaN	891	2
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male
freq	NaN	NaN	NaN	1	577
mean	446.000000	0.383838	2.308642	NaN	NaN
std	257.353842	0.486592	0.836071	NaN	NaN
min	1.000000	0.000000	1.000000	NaN	NaN
25%	223.500000	0.000000	2.000000	NaN	NaN
50%	446.000000	0.000000	3.000000	NaN	NaN
75%	668.500000	1.000000	3.000000	NaN	NaN
max	891.000000	1.000000	3.000000	NaN	NaN

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
count	714.000000	891.000000	891.000000	891	891.000000	204	889
unique	NaN	NaN	NaN	681	NaN	147	3
top	NaN	NaN	NaN	347082	NaN	G6	S
freq	NaN	NaN	NaN	7	NaN	4	644
mean	29.699118	0.523008	0.381594	NaN	32.204208	NaN	NaN
std	14.526497	1.102743	0.806057	NaN	49.693429	NaN	NaN
min	0.420000	0.000000	0.000000	NaN	0.000000	NaN	NaN
25%	20.125000	0.000000	0.000000	NaN	7.910400	NaN	NaN
50%	28.000000	0.000000	0.000000	NaN	14.454200	NaN	NaN
75%	38.000000	1.000000	0.000000	NaN	31.000000	NaN	NaN
max	80.000000	8.000000	6.000000	NaN	512.329200	NaN	NaN

Observation:

- The dataset contains 891 rows and 12 columns.
- Columns like `Age`, `Cabin`, and `Embarked` have missing values.
- Most passengers traveled in 3rd class (`Pclass = 3`).
- Several columns are categorical (e.g., `Name`, `Sex`, `Embarked`).

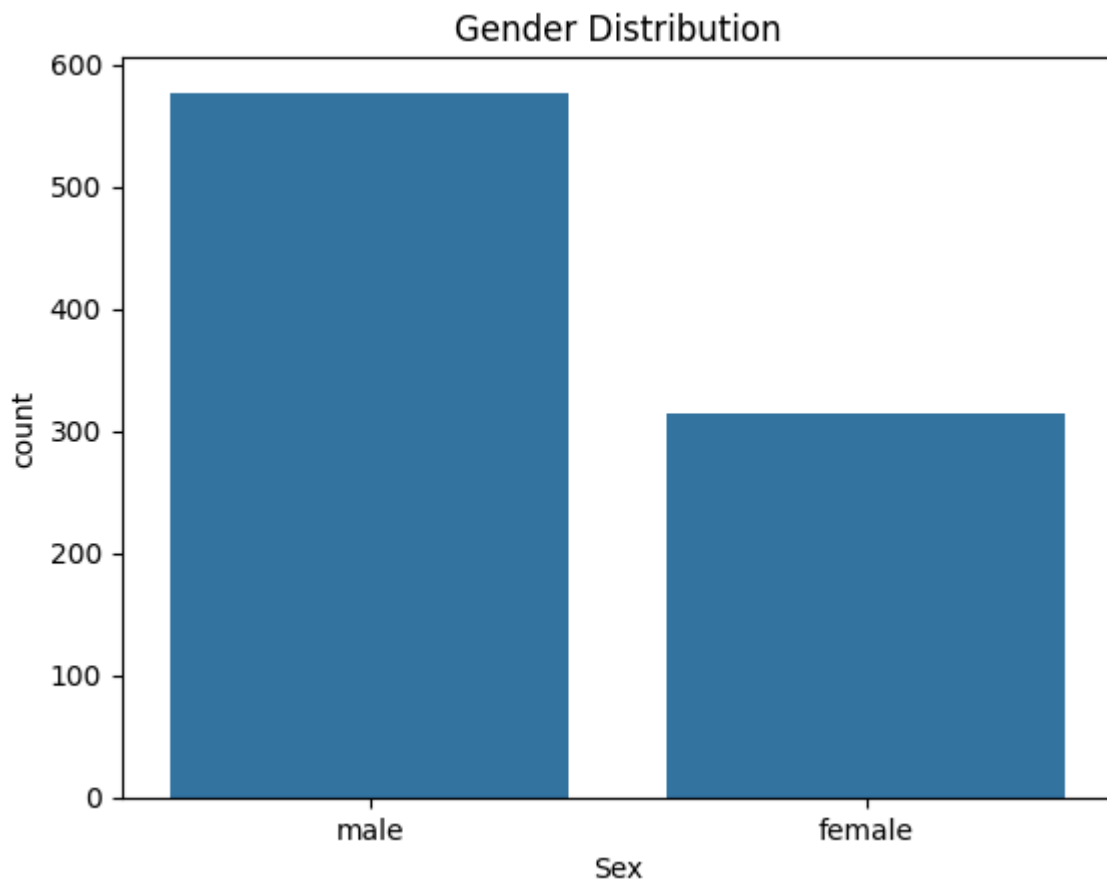
```
In [4]: print("\nMissing values:\n", df.isnull().sum())
```

```
Missing values:
 PassengerId      0
 Survived         0
 Pclass          0
 Name            0
 Sex             0
 Age            177
 SibSp           0
 Parch           0
 Ticket          0
 Fare            0
 Cabin          687
 Embarked        2
dtype: int64
```

Observation:

- Age has 177 missing values.
- Cabin has a large number of missing values (687).
- Embarked has 2 missing values.
- These columns may need to be cleaned or handled before modeling.

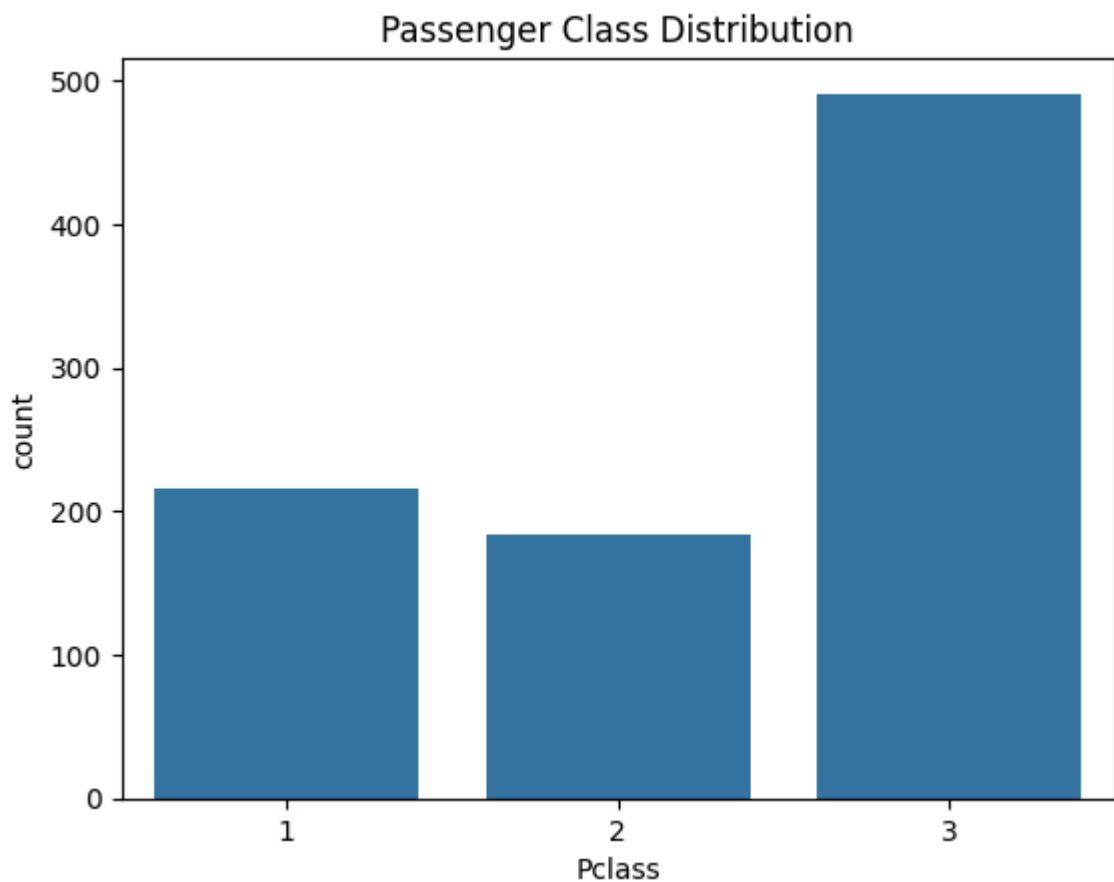
```
In [5]: sns.countplot(x='Sex', data=df)
plt.title("Gender Distribution")
plt.show()
```



Observation:

There were more male passengers than female on the Titanic. This gender imbalance may affect survival analysis.

```
In [12]: sns.countplot(x='Pclass', data=df)
plt.title("Passenger Class Distribution")
plt.show()
```



Observation:

Most passengers were in 3rd class, followed by 1st and 2nd. This shows a majority of low-fare travelers.



Summary of Findings:

- The majority of passengers were male and in 3rd class.
- Most passengers were aged between 20 to 40 years.
- There were more people who died than survived.
- Females had a higher survival rate than males.
- Passengers in 1st class had better chances of survival.
- Missing values are present in the `Age`, `Cabin`, and `Embarked` columns.
- `Pclass`, `Sex`, and `Fare` appear to be related to survival.